

Introduction to Sustainability

Sustainable Communication & Information
Technology



OUTLINE

- ❖ What is Sustainable Communication?
- ❖ How is Sustainable Communication different from ESG Communication?
- ❖ ICT and Environmental Sustainability
- ❖ Green IT: Saving Energy With Cloud Computing
- ❖ Practical implementation of Green
- ❖ Innovations from a micro-macro prospective



1. What is Sustainable Communication?



"Sustainability Communication is a strategic approach to engage stakeholders who either are looking up to you as consumers, or institutional investors with sustainability portfolio, willing to invest in you. It is extremely important for all stakeholders, but specifically for investors to know that you are truly on a sustainable journey"



2. How is Sustainable Communication different from ESG Communication?

The answer depends on who are you engaging with.

While Sustainable communication caters many stakeholders, including customers and employees, ESG communication is mainly meant to engage investors.

Sustainable Communication & ESG Communication

Sustainable Communication

Sustainable communication is a generic term to highlight a **sustainability strategy of the company**.

It tells customers, consumers and other stakeholders that a company is committed to pursuing sustainable development and is moving towards a more sustainable transition.

This role usually sits with the sustainability leadership or **corporate social responsibility teams (CSR)** within a company. And the primary stakeholders are end customers and investors.

ESG Communication

The target audience for **ESG (Environment, Social or Governance factors)** communication is very much Investors.

ESG is not a common term used among the consumers but is largely driven by Institutional investors who regularly look for ESG Issues that a company may have.

Investors assess ESG factors, specifically those which have a material impact on a company.

This role is very largely “Investor relations” with a collaboration with corporate social responsibility teams and sustainability leadership.

Do you pursue Sustainability Communication, ESG Communication or both?



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3. ICT and Environmental Sustainability

The promise of a decreased environmental footprint because of paperless offices and teleworking has been opposed by claims of increased power consumption and hazardous waste because of ICT.

The environmental footprint of ICT can be considered on different levels.

The First Level

Environmental effect because of the physical existence of ICT, causing increased consumption of energy, increased emission of greenhouse gas and increased nonrecycled solid waste.

The Second Level

Environmental effects because of the use and application of ICT.

On this level effects are generated by virtual goods, virtual stores, tele working, tele meetings, tele collaboration etc.

The Third Level

Systemic effects of the use of ICT.

These effects include impacts on facilities managed, on supply chains, on total freight transport and on total passenger transport.

An aerial photograph of a modern architectural complex. The central feature is a large, rectangular green roof with rows of plants. To the left, a curved, white-walled structure with a glass railing overlooks a lush courtyard filled with various trees and shrubs. A paved walkway with a wooden bench and a small white utility box is visible. The building's facade is a mix of light-colored panels and dark window frames. The entire scene is framed by stylized, wispy clouds in shades of blue and white, giving it a dreamlike or ethereal quality.

What is the role of ICT in
Sustainability?

ICT and Sustainability

- ◆ ICT and social sustainability
- ◆ ICT is pervasive and cross cutting
- ◆ ICT is a key enabler in the creation of networks
- ◆ ICT fosters the dissemination of information and knowledge
- ◆ ICT can radically reduce transaction costs
- ◆ ICT can enhance efficiency
- ◆ ICT enables innovation
- ◆ ICT facilitates disintermediation

4.

Green IT: Saving Energy With Cloud Computing

Energy can also be saved through cloud computing, namely the principle of outsourcing the programmes and functions of one's own computer to service providers over the internet.



GREEN IT: SAVING ENERGY WITH CLOUD COMPUTING

- ◆ Many individuals, governments and organizations are working to find ways to reduce energy demand in a smarter way, encouraging new ideas for using existing and new IT resources. One way to accomplish this is through cloud computing.
- ◆ Here are a few ways that moving to the cloud can help reduce the energy burden of IT services.
 1. Optimizing hardware resources
 2. Using energy efficiently
 3. Reducing costs
 4. Reducing data center footprints

TOWARDS ENERGY-EFFICIENT CLOUD COMPUTING

- ♦ Energy-aware data centres
- ♦ Energy savings in networks and protocols
- ♦ The effect of Internet applications



5. Practical Implementation of Green IT

Nowadays in order to achieve social awareness and promotion of green technology solutions, main four complementary approaches are employed:

- ♦ **Green Use:**

Reducing the power consumption of computers, information systems and their peripheral subsystems in environmentally friendly manner.

- ♦ **Green Disposal:**

Refurbishing and reusing existing old computers and other electronic associated devices.

Recycling unwanted used computers and other electronic-waste by IT vendors using their “take back” policy in order to take responsibility for the full lifecycle of products they produce.

♦ Green Design:

In broader aspect connecting companies, government agencies and environmental organizations in order to develop inventive management, business and regulatory processes that can improve environmental quality while enhancing economic development.

In narrow and practical aspect designing power efficient and eco-friendly computers and its subsystems like servers and cooling equipment.

♦ Green Manufacturing:

Process of production of computers and associated devices include methods of manufacturing and biodegradable components for minimal or no impact on environment.

This approach allows to provide economic benefits like long-term cost savings, and business process efficiency improvements.



6. Innovations from a micro-macro perspective

Latest Developments

- ♦ Green Cloud Computing
- ♦ Carbon aware green cloud architecture
- ♦ Data Center sustainability improvements
- ♦ Solar Computing
- ♦ Telecommuting



Thank
You